DEPI | DATA ANALYSIS TRACK

Project Planning & Management

Project proposal, schedule, task assignment, risk assessment, and KPIs

Air Quality and Life Expectancy: Exploring the Relationship Between Air Pollution and Population Health

Team Members

Ganat Saad Abdullah Hussein Ahmed | **Team Leader**
Shaimaa Maghraby Mohamed Houssien Elagouz
Habiba Maged Abdel Baqi
Dina Farag Mohamed Mohamed
Nada Ramzy Mohamed Ghnam

Prepared: 15 October 2026 | Final project target: 30 November 2026

1. Project Proposal

1.1 Project Overview

This project explores the relationship between ambient air pollution and population health by combining city-level air-quality observations with country-year health indicators. The analysis focuses on PM2.5, PM10, life expectancy, and adult mortality, with selected socioeconomic variables used where the available data support comparison.

1.2 Problem Statement

The selected datasets differ in geographic level and time coverage. Air-quality observations are recorded for cities, while life-expectancy data are recorded by country and year. The data therefore need to be checked, standardized, aggregated, and matched before the team can explore whether recorded particulate-matter levels are associated with differences in life expectancy and adult mortality. Missing values and uneven monitoring coverage may limit the comparisons.

1.3 Project Objectives

- Inspect and prepare both datasets, documenting relevant data-quality issues.
- Aggregate available city-level PM2.5 and PM10 observations to country-year level and retain the number of cities contributing to each average.
- Match air-quality and life-expectancy records using standardized country identifiers and year.
- Explore associations between PM2.5/PM10 and life expectancy/adult mortality in the matched observations.
- Examine differences by development status and consider GDP and schooling where sufficient data are available.
- Communicate the analysis through documented Excel/Power Query work, SQL where used, and a Power BI dashboard.

1.4 Research Questions and Scope

1. Is average PM2.5 associated with life expectancy in the matched country-year observations?
2. Is average PM10 associated with adult mortality in the available matched records?
3. Do the observed patterns differ between Developed and Developing countries where enough records are available?
4. How do GDP and schooling relate to life expectancy in the matched sample?
5. How complete and geographically representative are the available air-quality observations?

Scope: This is an exploratory observational project. It can identify associations but cannot establish that pollution caused a particular health outcome. Country-year air-quality averages will summarize only the cities represented in the source file, not necessarily the full national exposure.

1.5 Datasets and Sources

Dataset	Observed structure in supplied file	Relevant fields and preparation
WHO Ambient Air Quality Database (2018-version workbook; Kaggle)	11,962 rows; 15 columns; 108 countries; 4,468 distinct cities; years 2008–2017.	Country, City, Year, PM10, PM2.5, temporal coverage, conversion notes, monitoring-station information. Clean numeric values carefully, preserve conversion notes, standardize country names, and aggregate city records by country-year.
Life Expectancy Data (Kaggle)	2,938 rows; 22 columns; 193 countries; years 2000–2015.	Country, Year, Status, Life expectancy, Adult Mortality, GDP, Population, Schooling and other health indicators. Check missing values, duplicates, data types, field names, and country names.

Dataset links:

- Life Expectancy Data: https://www.kaggle.com/datasets/kumarajarshi/life-expectancy-who
- Ambient Air Quality Database: https://www.kaggle.com/datasets/fkosmowski/ambient-air-quality-database

1.6 Tools and Analytical Methods

- Excel / Power Query: data profiling, cleaning, country-name mapping, aggregation, and quality checks.
- SQL: joins, grouped summaries, and validation queries if used by the team.
- Power BI: dashboard visuals, filters, and comparisons.

- Methods: descriptive statistics, missing-value checks, grouped averages, trend comparisons, scatter plots, and correlation analysis. Correlations will be interpreted as associations, not causation.

2. Project Plan & Gantt Schedule

The proposed schedule targets final project completion by 30 November 2026. The first-submission documentation deadline is 16 October 2026.

Task / Phase	Start	Finish	Duration	Main output
Review dataset structure and quality	15 Oct 2026	19 Oct 2026	5 days	Data dictionary and coverage/missingness checks
Clean and standardize source files	20 Oct 2026	27 Oct 2026	8 days	Clean fields, conversion notes, country-name mapping
Aggregate air quality and merge	28 Oct 2026	3 Nov 2026	7 days	Country-year table and merge checks
Exploratory analysis and KPI calculations	4 Nov 2026	10 Nov 2026	7 days	Summary tables and descriptive analysis
Excel/SQL analysis and validation	11 Nov 2026	16 Nov 2026	6 days	Analysis workbook/queries and validation
Build Power BI dashboard	17 Nov 2026	23 Nov 2026	7 days	Dashboard with relevant filters
Review results and limitations	24 Nov 2026	26 Nov 2026	3 days	Checked results and documented limitations
Documentation and final presentation	27 Nov 2026	30 Nov 2026	4 days	Final report, presentation, and deliverables

3. Task Assignment & Roles

The role allocation below is preliminary and may be adjusted as the project progresses. Each member has a main responsibility, while collaboration across roles is expected.

Role	Assigned member name	Primary responsibility	Tasks / deliverables
Role 1	Ganat Saad Abdullah Hussein Ahmed | Team Leader	Project Coordinator & Excel Data Preparation Analyst	Coordinate project activities, organize tasks, and monitor the project schedule; maintain project documentation; inspect the source datasets in Excel; check missing values, duplicates, inconsistent country names, and other data-quality issues; help define analytical questions and review the team's outputs.
Role 2	Habiba Maged Abdel Baqi	Data Transformation & Integration Analyst	Use Power Query in Power BI to transform and integrate the datasets; aggregate city-level air-quality observations by country and year; standardize country names and prepare matching fields; validate the merged dataset and document unmatched records and coverage limitations.
Role 3	Shaimaa Maghraby Mohamed Houssien Elagouz	Exploratory Data Analysis (EDA) Analyst	Perform descriptive analysis of PM2.5, PM10, life expectancy, and adult mortality; explore trends and potential associations in matched records; compare countries and years where data are available; identify patterns, unusual values, and limitations.
Role 4	Dina Farag Mohamed Mohamed	Data Visualization & Dashboard Analyst	Develop suitable charts and the Power BI dashboard; visualize trends, comparisons, and potential relationships; check that visuals accurately represent the validated data; coordinate with the EDA role to ensure visualizations address the project questions.
Role 5	Nada Ramzy Mohamed Ghnam	Research, Reporting & Presentation Analyst	Review credible sources and maintain references; summarize how sources inform project variables and interpretation; contribute to interpreting results with the team; help organize the final project report and presentation materials; participate in developing and reviewing the Power BI dashboard and its visuals.

4. Risk Assessment & Mitigation Plan

The team will monitor the following project-specific risks and apply the listed mitigation actions.

Risk	Potential impact	Mitigation plan
Missing health or pollution values	Reduced sample size or misleading summaries.	Measure missingness first, document exclusions, and avoid unsupported imputation.
Country-name mismatches	Valid country-year observations may fail to match.	Use reliable ISO-3 codes where available and maintain a checked country-name mapping.
PM values contain conversion notes or text	Numeric calculations may fail or mix values with different origins.	Preserve raw values and conversion notes; create clean numeric fields and flag converted observations.
Uneven city and year	Country averages may not be	Retain city counts and coverage information; describe averages as summaries

coverage	comparable or representative.	of available cities and avoid overgeneralizing sparse years.
Association interpreted as causation	Conclusions may claim more than the observational data support.	Use association language and state relevant data limitations and possible confounding factors.

5. Key Performance Indicators (KPIs)

The following indicators are proposed for describing the data and monitoring the analytical coverage. Their values will be calculated from the cleaned, matched data.

KPI	Definition / calculation	Importance
Average PM2.5 (μg/m^3)	Mean of valid PM2.5 observations within each country-year after documented cleaning.	Main fine-particle pollution measure.
Average PM10 (μg/m^3)	Mean of valid PM10 observations within each country-year after documented cleaning.	Second particulate-matter measure for comparison.
Life expectancy (years)	Life-expectancy value from the source dataset for each matched country-year.	Primary population-health outcome.
Adult Mortality	Adult Mortality value as defined in the source dataset.	Secondary health outcome; interpret using source definition and units.
City coverage count	Number of distinct cities contributing valid observations to each country-year pollutant average.	Shows the amount of city-level information supporting each average.
Matched country-year count	Number of unique country-year keys present in both prepared datasets.	Measures the final integration coverage and analytical sample size.
Data completeness (%)	Non-missing valid values divided by expected values, multiplied by 100, calculated for key fields.	Shows how much of the required data is available.
Pollution-health correlation	Pearson correlation between a pollutant measure and a health outcome using valid matched pairs.	Summarizes direction and strength of linear association; it does not establish causation.